Department of Computer Engineering and Informatics

Faculty of Science and Technology, Middlesex University Dubai



CST2120

Web Applications and Databases

Coursework 1: Report

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Repository Link: https://github.com/JaferDev/Jafer-TCG

Video Link:

INDEX

1. Introduction 2
1.1. Project Overview and Game Idea 2
1.2. Aim and Objectives 3
1.3. Scope 3
2. Planning and Requirement Analysis 3
2.1. Game Concept 3
2.2. Functional Requirements 3
2.3. Non-Functional Requirements 3
2.4. Hardware and Software Requirements: 4
3. System Design 4
3.1. Website Navigation 5
3.2. Wireframes and UI Design 5
3.3. Game Mechanics and Rules 6
3.3.1. Selecting Characters: 6
3.3.2. Selecting Energy: 6
3.3.3. Selecting Active Character: 7
3.3.4. Turn Based Combat: 7
3.4. Game Flowchart 7
..... 8
3.5. Registration System Design 8
3.6. Login System Design 8
4. Development and Implementation 9
4.1. Website Structure 9
4.2. Player Controls 9
4.3. Registration and Authentication 9
4.4. Score, Leaderboard, and Rankings 12
4.5. Advanced Game Features 12
5. HTML and CSS validation 16

6. Evaluations and Future Improvements 20
6.1. Achievement of Project Objectives 21
6.2. Limitations and Challenges 21
6.3. Future Scope and Improvements 21
7. Conclusion 21
8. Bibliography 22
9. AI Disclosure 24

1. Introduction

1.1. Project Overview and Game Idea

This project was created as a submission for CST 2120 Coursework 1. The project is a website and a video game created using HTML5, CSS, and JavaScript. All front-end was done using only the three languages mentioned and the back-end was done using HTML5's local storage and session storage.

My game idea was to create a turn-based battle video game similar to Pokémon's 'Pokémon TCG' and Genshin Impact's 'Genius Invocation TCG' with unique characters with their own attacks.

1.2. Aim and Objectives

The aim of this project was to successfully create the game in an efficient and effective manner, taking inspiration from the Unity Game engine to design the graphics display and event handling systems. I also aimed to have all code written by me and all art drawn by me.

1.3. Scope

This game is meant to be a simple and small version of the games mentioned earlier. It took approximately 7 weeks to create and was made entirely by one individual.

2. Planning and Requirement Analysis

2.1. Game Concept

The game website and game will be created using HTML5, CSS, and JavaScript. For the actual game, HTML5's Canvas API will be used. It will be a turn-based battle game with many unique characters with their own character cards and attacks. All character cards will be rendered using HTML5 and CSS.

2.2. Functional Requirements

1. The player can start the game
2. The player can login, register, or play as a guest
3. The player can select a hand from a selection of character cards
4. Game keeps track of the score, time played, and the high score
5. High score will be saved for all players logged in with a score greater than the current high Score

6. The player moves onto a new opponent with new hand after defeating current opponent

2.3. Non-Functional Requirements

1. The game should be responsive
2. It should be simple to understand the controls and the tutorial
3. The character cards should display important information in a readable manner
4. The game should run without any errors on Google Chrome

2.4. Hardware and Software Requirements:

- Hardware: Windows, Mac, or Linux OS. 100MB in storage
- Software Requirements: Google Chrome

2.5. Gantt Chart

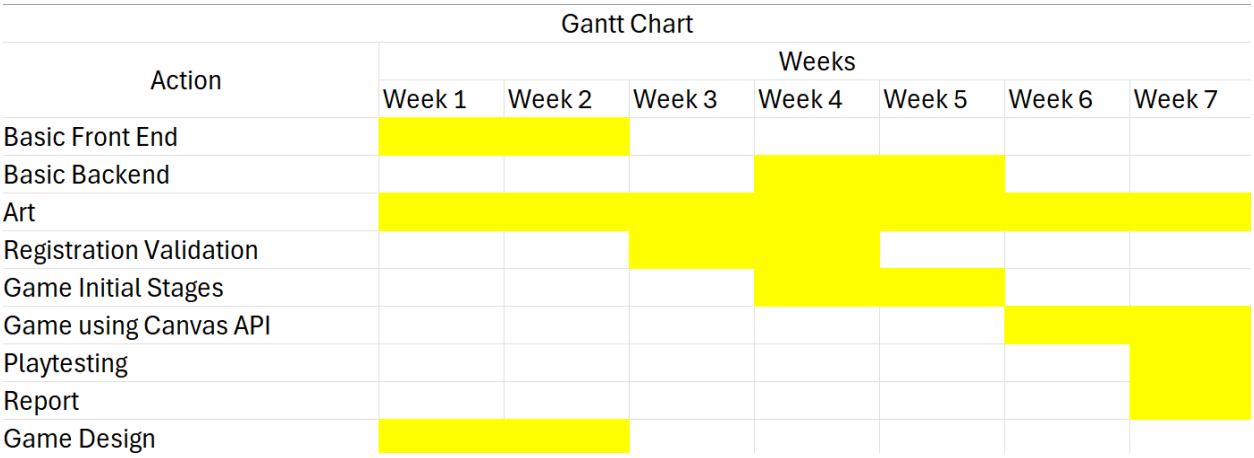


Fig. 1 Showcasing a Gantt Chart plan created in week 1

3. System Design

3.1. Website Navigation

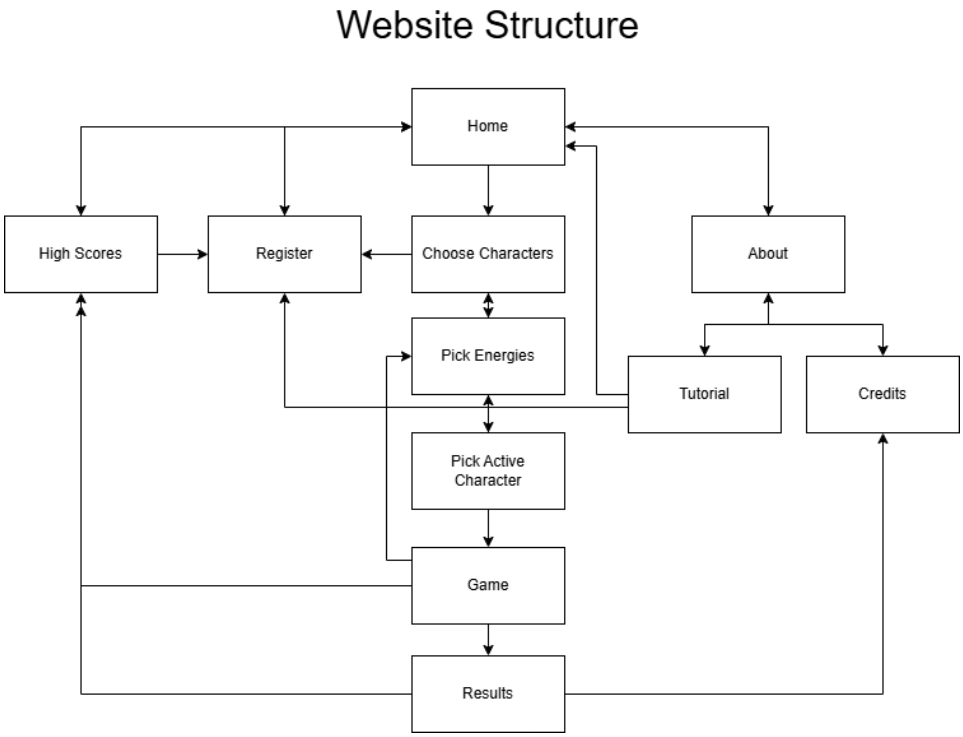


Fig.2. Navigation structure of website

3.2. Wireframes and UI Design

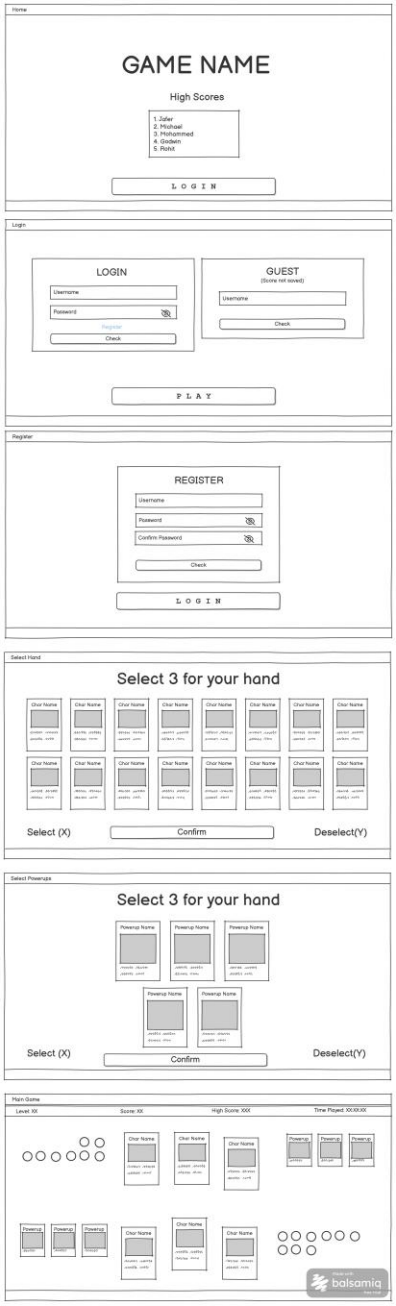


Fig.3. Initial wireframe idea created at week 1

3.3. Game Mechanics and Rules

3.3.1. Selecting Characters:

Six character cards are displayed on the screen, and the player must select three to play with.

3.3.2. Selecting Energy:

Ten energy tokens are shown face down, the player must click on any five at random, these will be the energies the player will play with.

3.3.3. Selecting Active Character:

The three Selected characters are displayed, and the player must select one of them to be their active character.

3.3.4. Turn Based Combat:

Player goes first
Player can either switch character, attack, or end round

- **Switch Character:**
  Requires any one energy and ends turn
- **Attack:**
  Requires the amount of energy shown in character card. Pick between two attacks. Reduces the HP of the enemy's active character.
- **End Round:**
  Ends player's turn immediately

The enemy goes next and choice is picked by a random number generator. The chance of each choice depends on the amount of energy the enemy has.

3.4. Game Flowchart

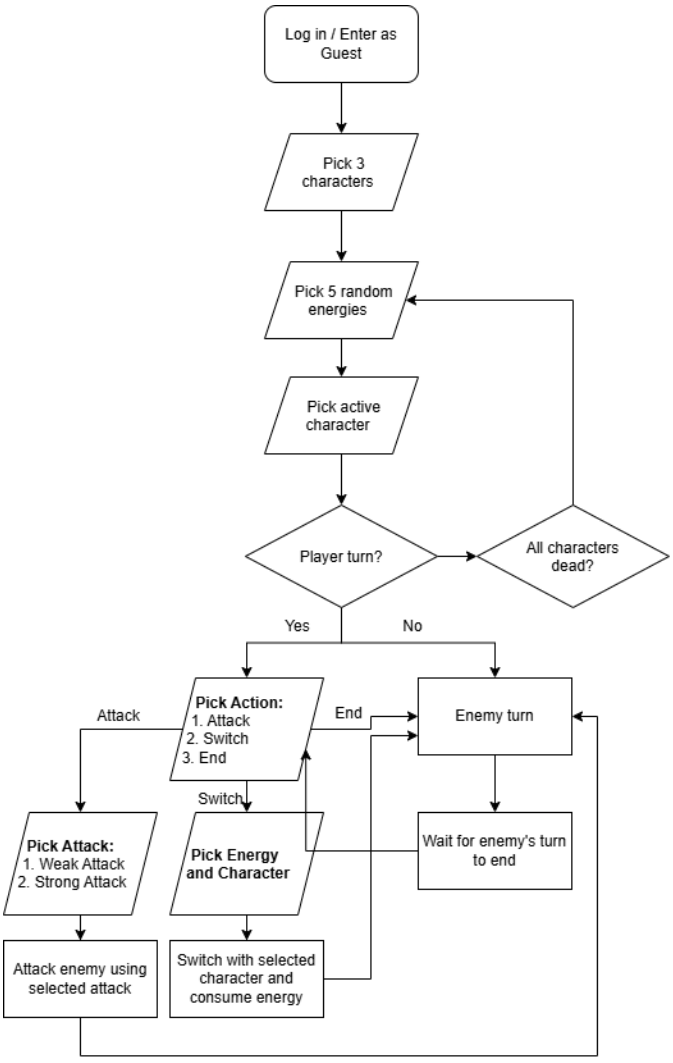


Fig.4. Game flowchart explaining game logic from the time the player logs in or enters as guest till the end of round

3.5. Registration System Design

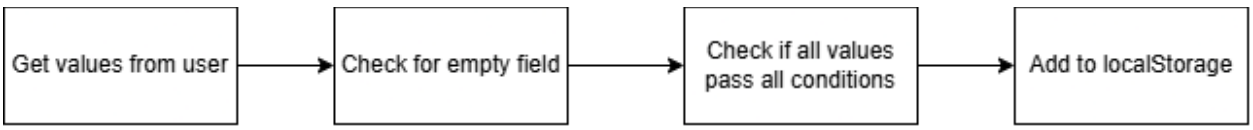


Fig.5. Registration System Design flowchart

If all the conditions are fulfilled, the registration manager adds the new user object to localstorage

3.6. Login System Design

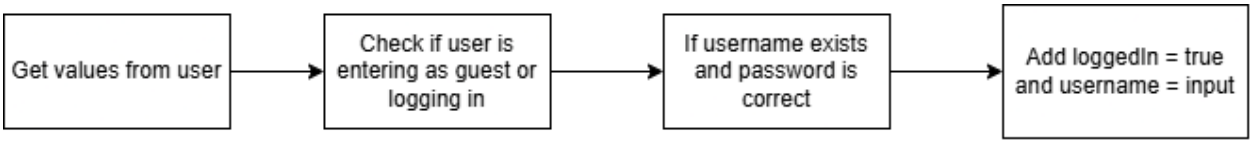


Fig.6. Login System Design flowchart

If the user exists and the password entered is correct, the user logs in and is added to the session storage.

4. Development and Implementation

4.1. Website Structure

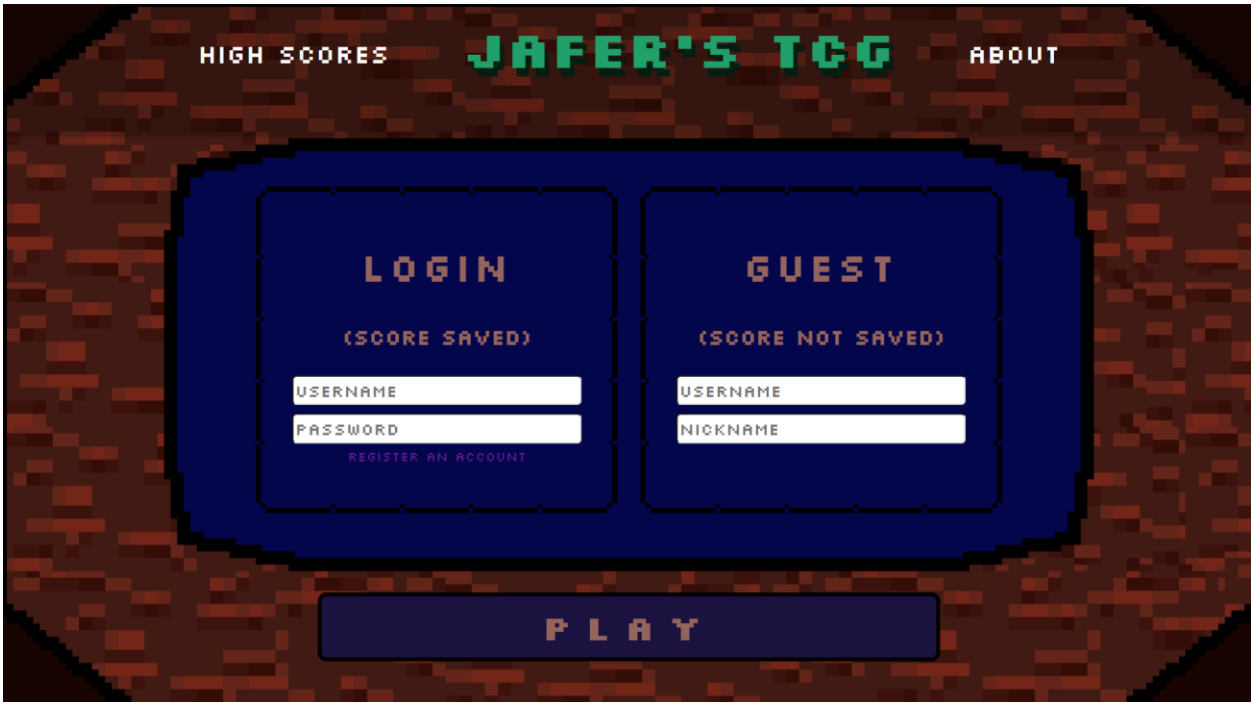


Fig.7. Game index.html screenshot showcasing the website structure with a horizontal navbar on the top, page head in the centre, div containers, and a long button on the bottom

4.2. Player Controls

The game is controlled using mouse, there are no keyboard controls except for entering login and registration details.

4.3. Registration and Authentication



Fig.8. Error showing "Fill all empty fields"



Fig.9. Errors showing for email since it should contain "." and "@"



Fig.10. Error showing "Confirm password should be same as password" and others for password since password entered does not have any special characters or numbers and is less than 8 characters



Fig.11. Error showing "Confirm password should be same as password" and "Age must be greater than 5" since dob was set as 2026

```
//Dictionary[str, bool] . Key is error message, value is condition required to fail to raise error message
const LOGINCONDITIONS = {
    "Email should include \".\"": email.includes("."),
    "Email should include \"@\"": email.includes("@"),
    "Email cannot include space": !email.includes(" "),
    "Email cannot be longer than 64 characters in total": email.length < 64,
    "Username exists": sessionStorage.getItem(username) !== null,
    "Username cannot include special characters": !CheckSpecialChars(username),
    "Username cannot be longer than 16 characters": username.length < 17,
    "Username must at least be 5 characters": username.length > 4,
    "You must be at least 5 years old": (CURDATE - dateOfBirth) / (1000 * 60 * 60 * 24 * 365) >= 5,
    "You cannot be more than 161 years old": (CURDATE - dateOfBirth) / (1000 * 60 * 60 * 24 * 365) <= 2391,
    "Password must be at least 8 characters": password.length >= 8,
    "Password must contain at least one special character": CheckSpecialChars(password),
    (property) "Password must contain at least one small letter": boolean
    "Password must contain at least one small letter": /[a-z]/.test(password),
    "Password cannot be more than 16 characters": password.length < 17,
    "Confirm password should be the same value as password": confPassword == password
}
```

Fig.12. List of all 16 conditions required to be fulfilled

4.4. Score, Leaderboard, and Rankings



Fig.13. Screenshot of 6_highscores.html, showing leaderboard with rankings and a score

4.5. Advanced Game Features

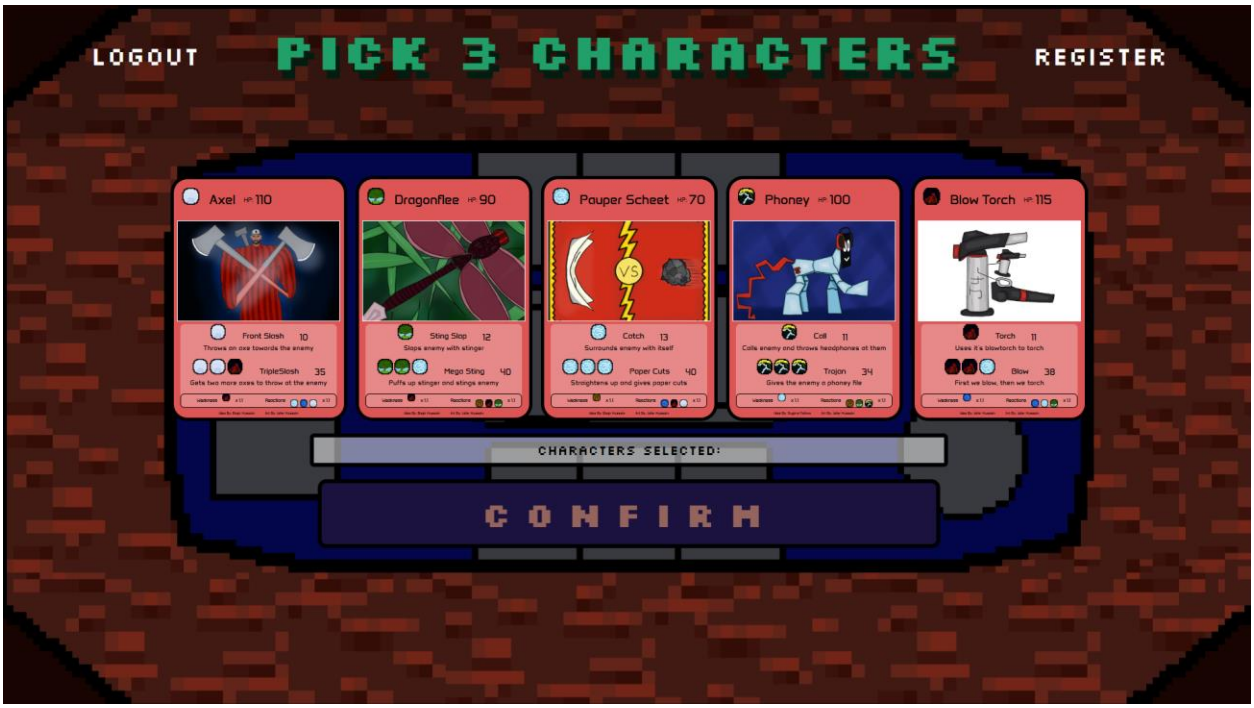


Fig.14. Player can pick 3 characters

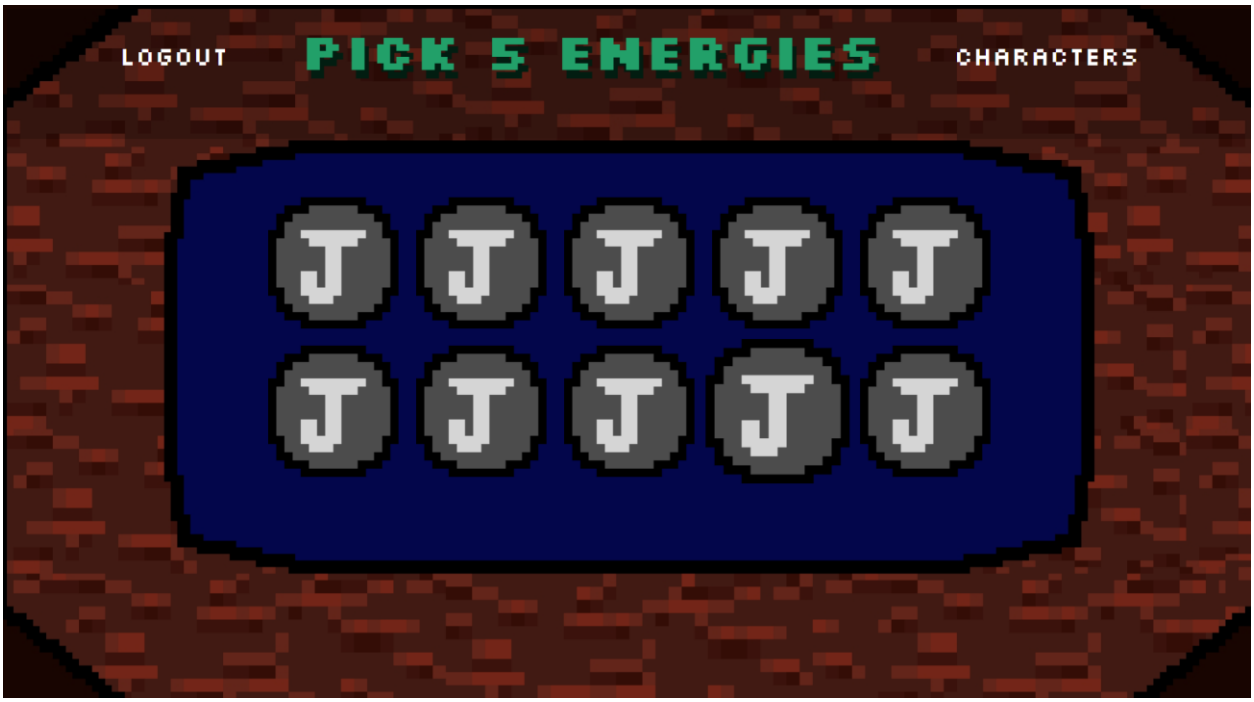


Fig.15. Player can pick any 5 energies



Fig.16. Sets a timer for 3 seconds to go to next page after 5 energies are selected



Fig.17. Player can pick 1 active character



Fig.18. Initial display of game.html. Enemy on the right(right) and player on the left(blue). Player energies on the top left, Player attacks on the left. Inactive characters on the bottom

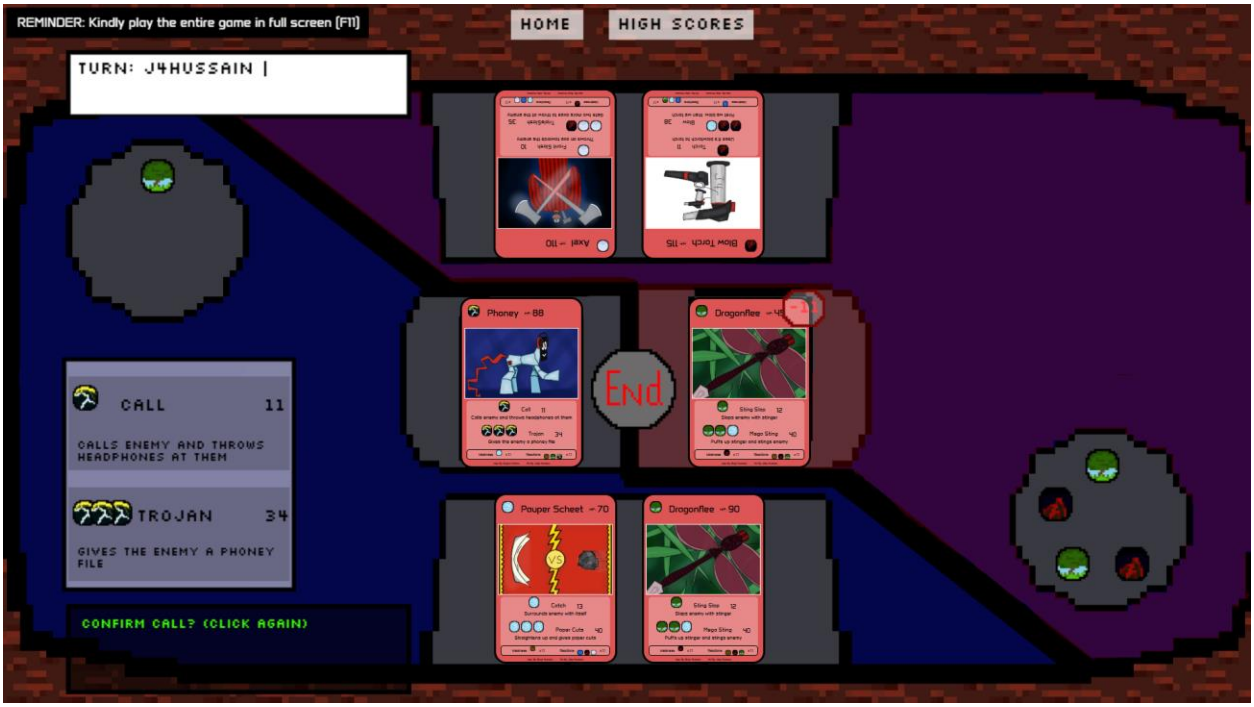


Fig.19. Player attacked enemy (See the circle with –11 beside enemy hp)

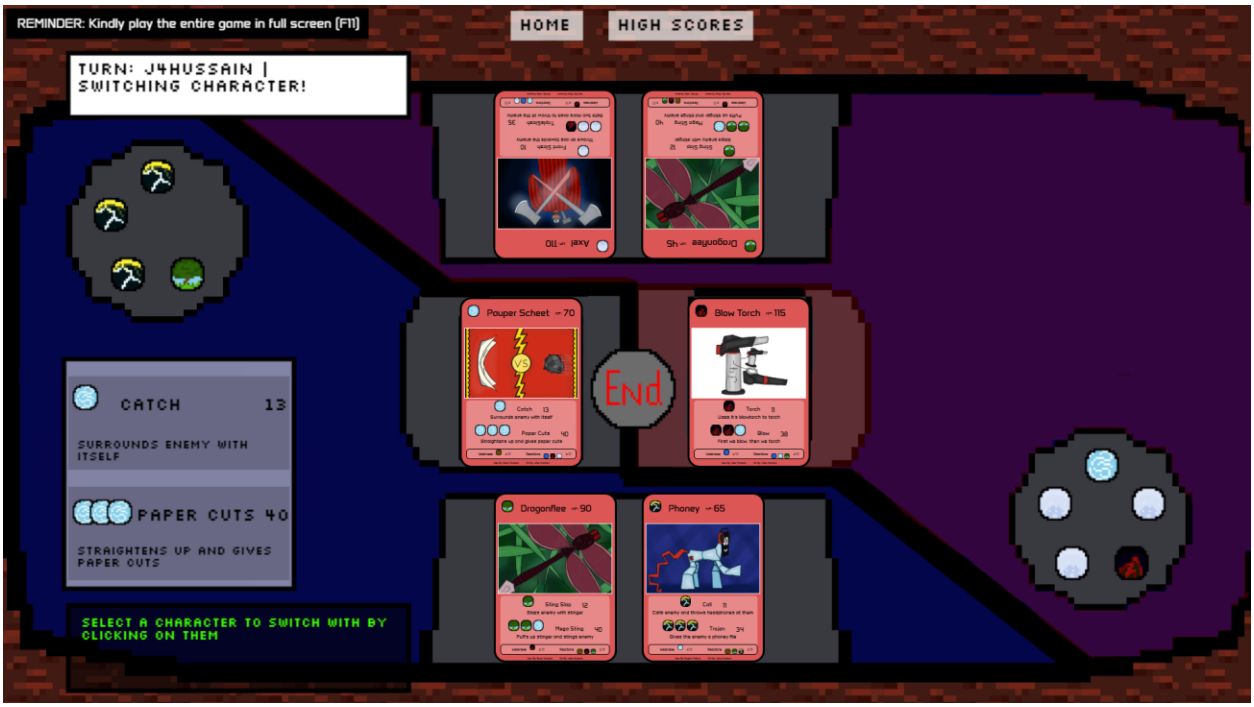


Fig.20. Player switched characters from Phoney to Pauper Scheet

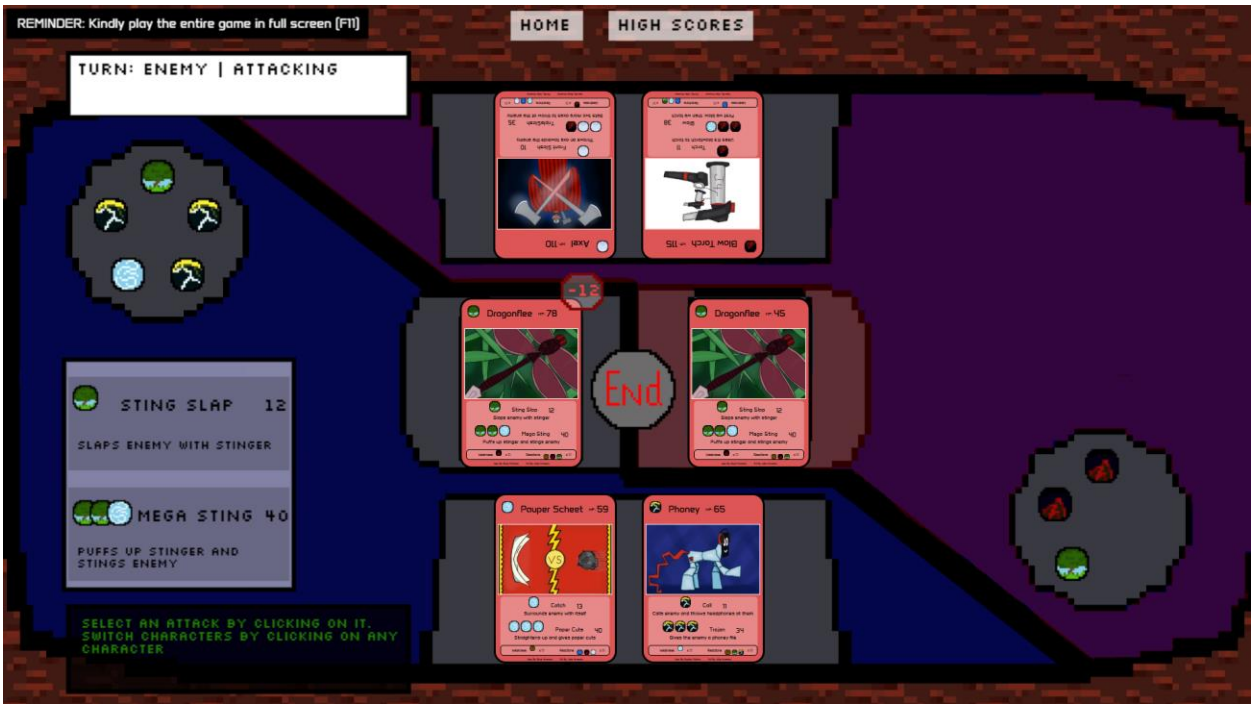


Fig.21. Enemy attacking player (See the –12 beside player hp)

5. HTML and CSS validation

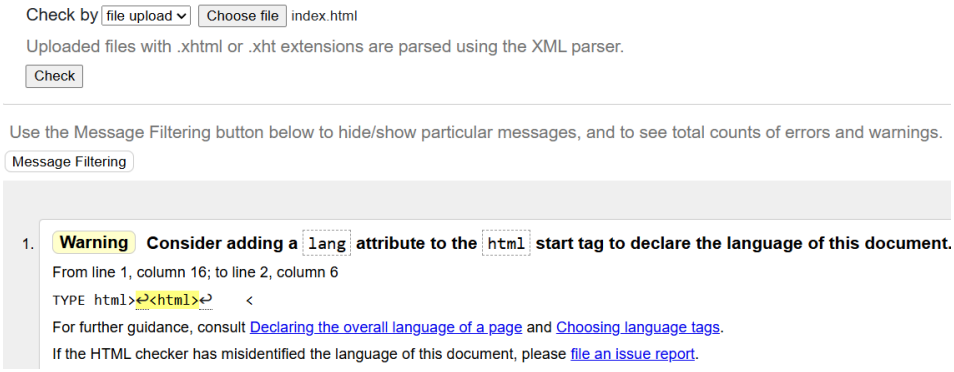


Fig.22. Validation of index.html



Fig.23.Validation of 1_register.html

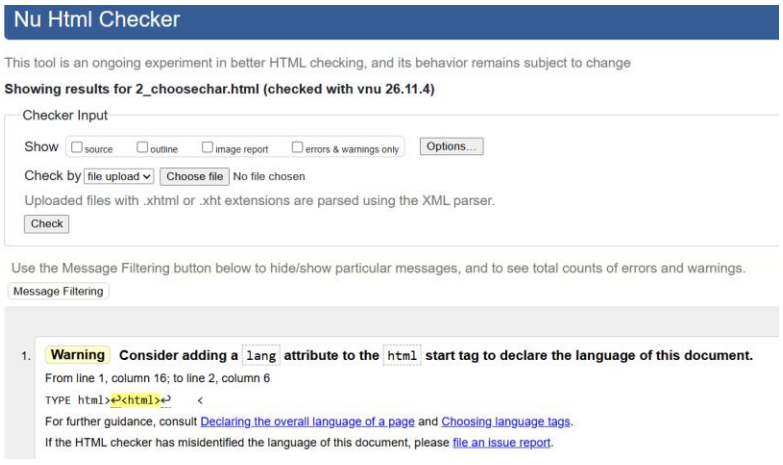


Fig.24.Validation of 2_choosechar.html

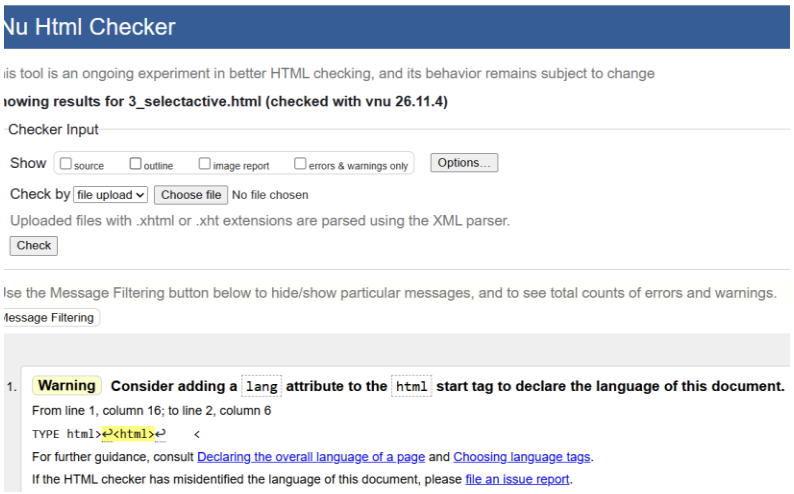


Fig.25.Validation of 3_selectactive.html

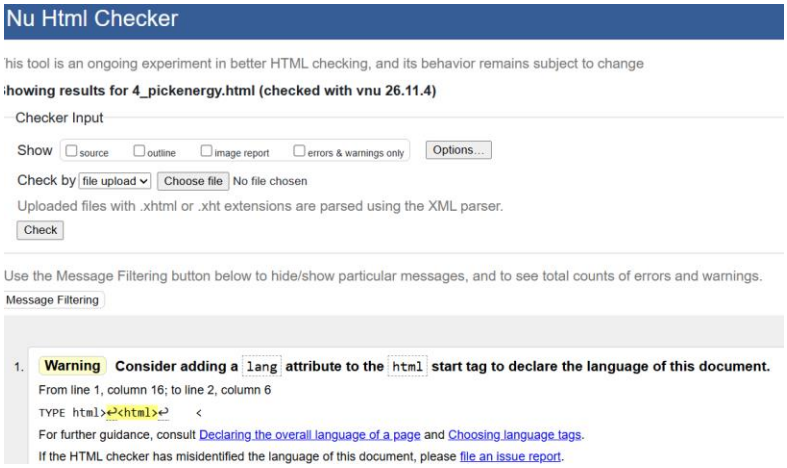


Fig.26.Validation of 4_pickenergy.html

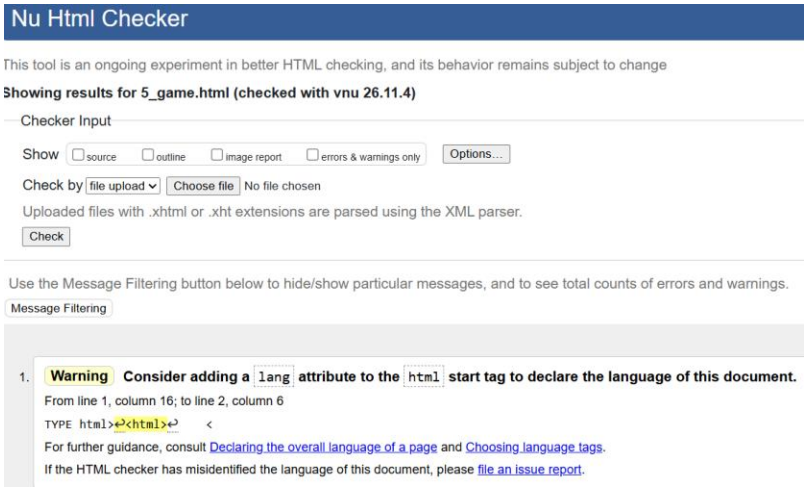


Fig.27.Validation of 5_game.html

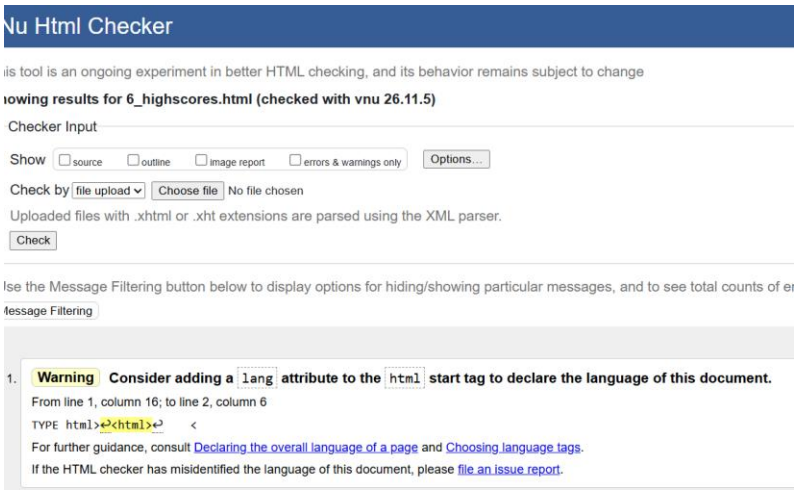


Fig.28.Validation of 6_highscores.html

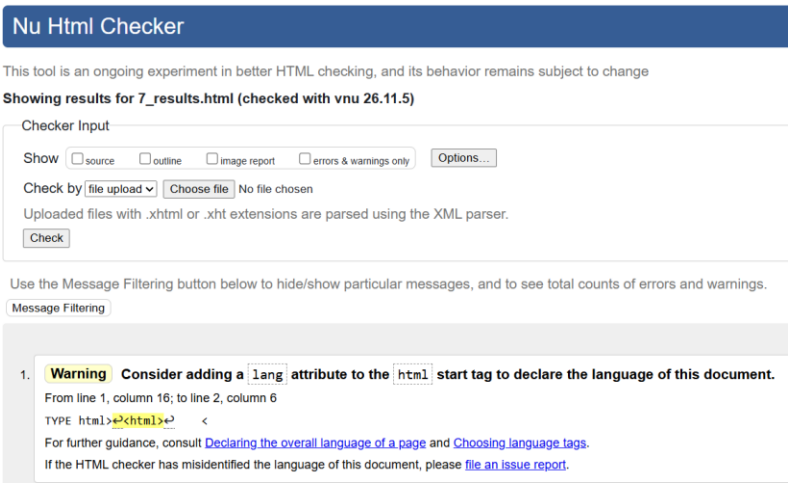


Fig.29.Validation of 7_results.html

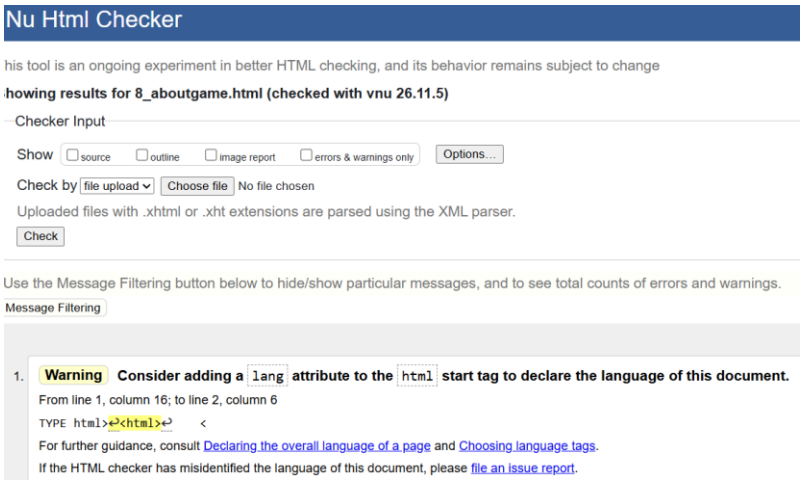


Fig.30.Validation of 8_aboutgame.html

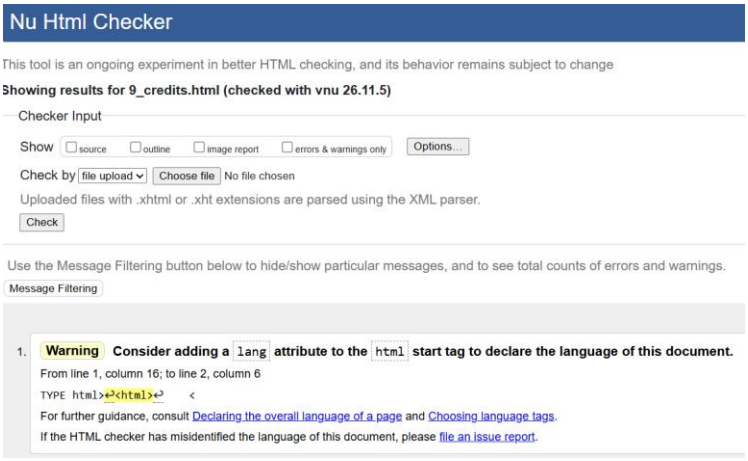


Fig.31.Validation of 9_credits.html

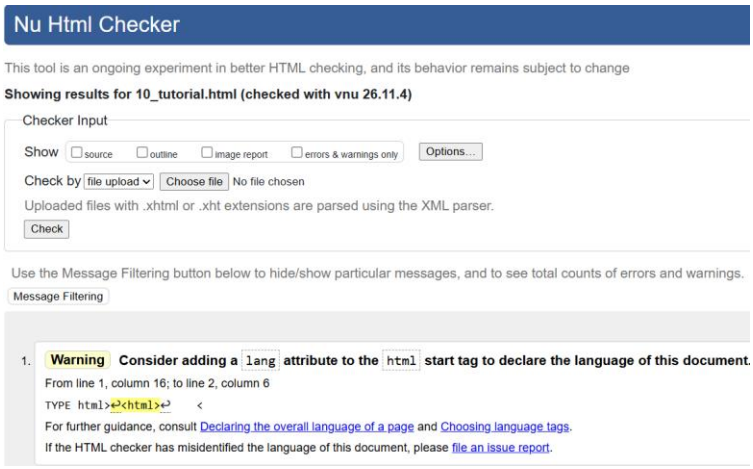


Fig.32.Validation of 10_tutorial.html

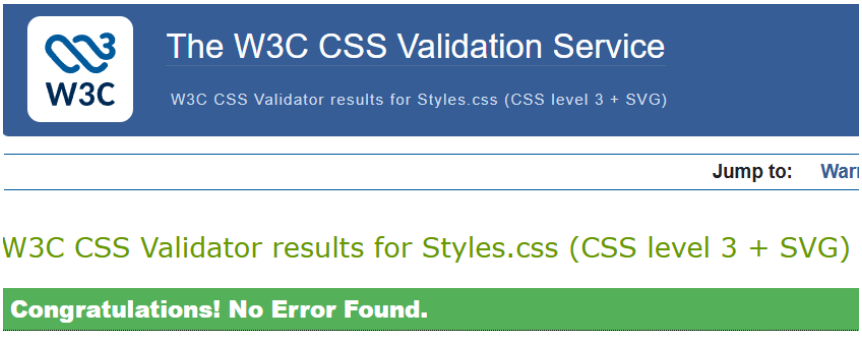


Fig.33.Validation of Styles.css

6. Evaluations and Future Improvements

6.1. Achievement of Project Objectives

I believe that I have done a really good job at creating an entire game and game website from scratch using JavaScript, HTML5, and CSS. I have achieved all my functional and non-functional objectives. I personally feel very accomplished and I'm sure all the lessons I've learnt will come to use in my future.

6.2. Limitations and Challenges

This coursework was one of the most challenging courseworks I have had the pleasure of completing. I didn't face much difficulty with the basic HTML and CSS, however displaying the character cards, creating the HTML5 Canvas game, and handling all the data that's necessary for the game to function was very difficult for me.

The first big challenge I had faced was displaying character cards. I had originally created them as images and displayed them as HTML <img> elements. However, I quickly realised that doing so would prevent me from showing any changes to the cards, for example when any character takes damage, I wouldn't be able to show the updated HP. So, I had to recreate my initial design entirely in HTML5 and CSS and create a function in JavaScript to dynamically display character cards.

Other than this, developing the game, learning HTML5's Canvas API, and OOP, also proved to be quite difficult.

6.3. Future Scope and Improvements

This was a very simple version of the games I was inspired with, so in the future I would like to make it significantly more complex, with improved game balancing and more complex damage calculations that take many more parameters into account.

The enemy I had designed was relatively simple since it decided it's actions using weighted probability. In the future I would like to apply a simple machine learning algorithm such as Decision Tree algorithms to improve the performance of the enemy.

I used HTML5's Canvas API instead of a game engine, although I don't regret my decision to use the Canvas API since I've learnt a lot, I believe to take the game to the next level it would be necessary to use a game engine.

7. Conclusion

To conclude this report, I believe I have done a great job displaying my newfound knowledge of HTML5, CSS, and JavaScript by developing a video game and a website for the game entirely by myself.

I have understood and implemented all the main concepts taught by my lecturer Dr. Sumitra Kotipalli. I greatly enjoyed this experience, and I can't wait to expand upon my current game and game website and take it to the next level.

8. Bibliography

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9. AI Disclosure

I have not used any generative AI to assist with my work on this project. This includes all code, this report, my video demonstration, and every sprite. The music has not been disclosed to have used any generative AI and has been sourced from Pixabay.